

Nishan Mudalige^{1,2,3}, Megan Milton¹ and Sujeevan Ratnasingham¹

¹ Biodiversity Institute of Ontario, University of Guelph, Guelph, ON, N1G 2W1, Canada. ² Department of Mathematics and Statistics, University of Guelph, Guelph, ON, N1G 2W1, Canada. ³ Ontario Graduate Scholar.

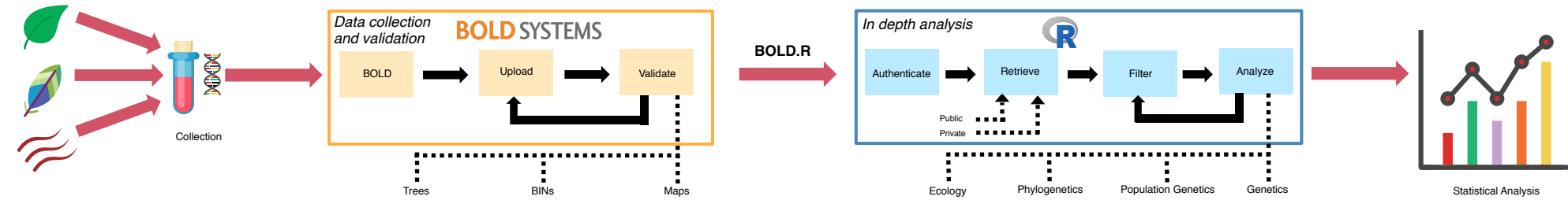


Figure 1: The pipeline showing the flow of information from collection in the field to statistical analysis.

Background:

The International Barcode of Life project (iBOL) continuously records and catalogs data collected from samples submitted from researchers around the world. This data is stored on the Barcode of Life Data system (BOLD). Advances in DNA analysis have led to a rapid increase in the volume of data available for researchers involved with the iBOL project to study. As a consequence, modern statistical techniques are playing an increasingly important role in the analysis of such large volumes of data. A popular software package for statistical analysis is R, however existing methods to enter data stored on BOLD into R are inconvenient, time-consuming or provide limited information. Therefore, we developed BOLD.R to provide a more accessible system to overcome these issues.

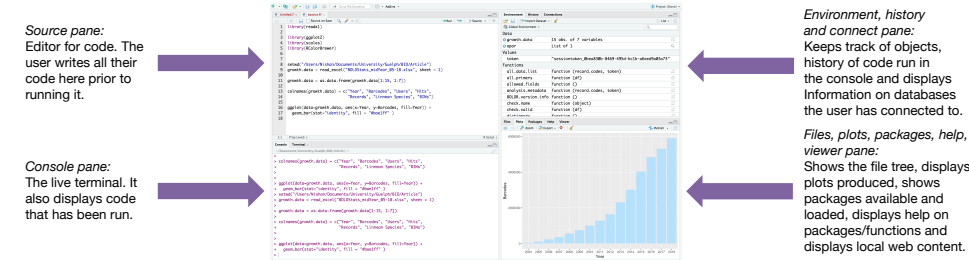


Figure 2: The interface of R-Studio which is a popular integrated development environment (IDE) for R. Note that the size and layout of panes can be modified.

Advantages of R

- Cost:** R is free open-source software.
- Cross-platform:** Available on all major platforms (Windows, Mac, Linux).
- Development environment:** R is built specifically for statistical analysis by statisticians and scientific researchers.
- Complete and Comprehensive:** The base version of R is well-equipped to perform standard statistical analysis (tests, model construction, etc.). Furthermore, there is an extensive selection of libraries developed to perform generic tasks, highly specialized tasks, and also for specific research areas.
- Visual:** R offers exceptional graphical capabilities and allows users to create professional plots for publishing.
- Efficient and Optimized:** The base version of R package is very efficient on resources and many libraries are often continuously updated until optimized.
- Format support:** R offers users support for many file formats (input and output).
- Community support:** Developers and researchers contribute to maintaining R and there is a large community of active users available for assistance.

Significance

In order to overcome existing barriers which prevent ease of access to information on BOLD directly into statistical software, we developed an R library called BOLD.R. This library allows users to load data directly from BOLD into R via current APIs maintained by the BOLD system. Users can access their own private data by logging in to BOLD using BOLD.R, or they can access public data without the need to login. All data accessed using BOLD.R is stored in R with a consistent internal structure and this allows the user to employ the suite of functions provided by BOLD.R along with existing packages to perform statistical analysis on biological data, including DNA sequences. BOLD.R is a powerful library which bridges the gap between data storage and data analysis.

Key field			Sequence Alignments			Custom fields	
Record code	Process id	API accessed	COL5P_nucraw	CYTB_nucraw	...	Extra info	Collectors
DS-GBONT1	GBAH10068-15	private	AAC...GGA	NA	...	taxon:1539396	CNC Hym team
DS-GBONT1	GBAH10069-15	private	-AC...GG-	NA	...	taxon:1539396	H. Goulet
...	...	...	...	...	...	...	...
DS-GBONT1	GBMIN14500-13	private	AAC...GA	NA	...	Bembidion affine	W. van Steenis
ACAGA	ASAG101-07	public	NA	ACA...ATA	...	NA	NA
ACAGA	ASAG137-07	public	NA	-CA...A-	...	NA	NA
...	...	...	...	...	...	...	...
ACAGA	ASAG137-07	public	-ATT...---	NA	...	NA	NA

Figure 3: Conceptual diagram of a data frame.

Advantages of Data Frames

Data retrieved using BOLD.R is stored as data frames. Data frames offer the following advantages

- Flexible:** Each column can store a different type of data (e.g. Strings, numeric, etc.), so we are not restricted to just one type of data in a data frame.
- Labelling:** Each column can be given a meaningful column name.
- Modifying:** The user can make changes to the data in a data frame without affecting the original data on BOLD.
- Filtering:** The user can filter data by values in a column.
- Merging:** Straightforward to merge data and remove duplicates.

An alpha version of BOLD.R is available for download at: <http://www.boldsystems.org/BOLD.R>

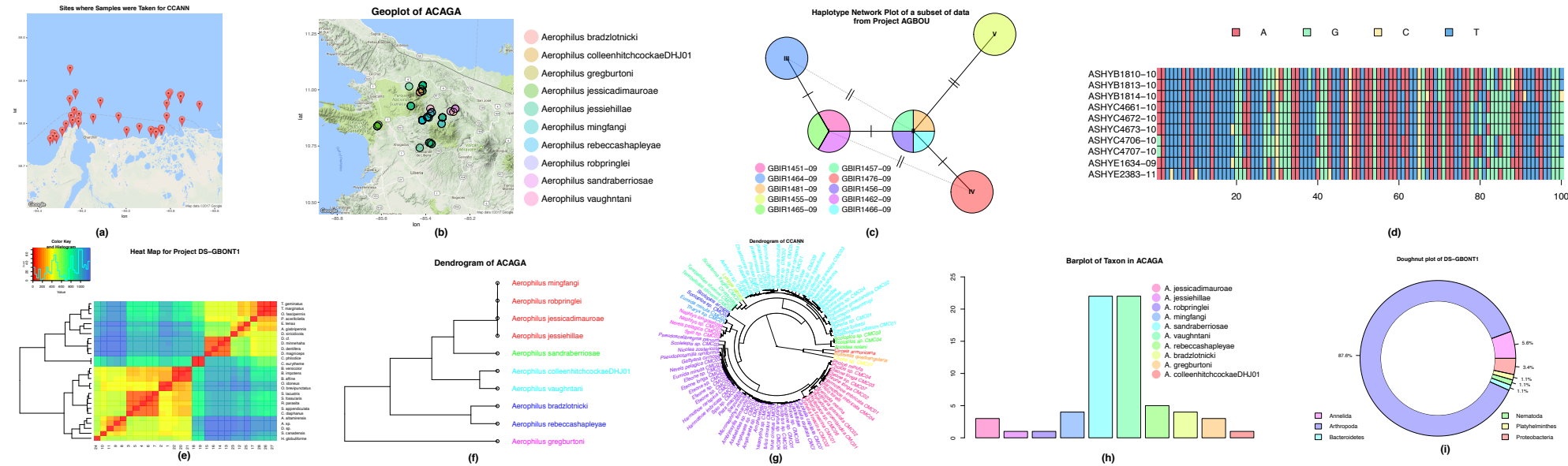


Figure 4: Various plots obtained using BOLD.R and existing R libraries. These examples illustrate visual representations of spatial distributions (a, b), networks (c), substitution (d), clustering (e), phylogeny (f, g) and composition (h, i). All data used to create these plots is publicly available on BOLD.

Note: All of the plots, maps and figures above were obtained using BOLD.R and existing R libraries.



Figure 5: Roadmap showing the integration of BOLD.R into a future implementation of BOLD.

Long Term Plan

- Employ the analytical engine of BOLD:** Develop the BOLD API and BOLD.R to harness the power of the analytical engine of BOLD through cloud computing.
- Establish a script repository:** Allow users to store their own scripts privately or make scripts public.
- Integrate popular scripts as Plugins on BOLD.**

Summary

BOLD.R is a powerful, flexible and convenient package which provides the user with the capacity to access and analyze large amounts of private and public data on BOLD directly through R. It can therefore become an invaluable tool to assist researchers make informed decisions through the automation of data processing and enable the analysis of extremely large datasets.

References

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